# THERE AND BACK AGAIN: A GENUS-THREE POLYHEDRON, ITS HYPERBOLIC TWIN, AND ITS RIGIDITY

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Abstract. In 2026 Mizhaev gave integer coordinates for an equivelar polyhedron $P$ of genus three, made of eight planar nonagons, every two of which share an edge. We pass from $P$ to a hyperbolic surface and back without artificial choices, and ask whether the way back into space is unique. Uniformisation turns $P$ into a hyperbolic surface $h_P$ with 24 marked points and cone angles; Troyanov's theorem inverts this, and numerically the round trip closes to $10^{-14}$. The combinatorial shortcut through circle packing is not canonical: its surface lies at Thurston distance at least 0.23 from $h_P$. Using exact integer arithmetic we prove that $P$ is infinitesimally rigid and that the edge graph of its triangulation is generically globally rigid in $\mathbb{R}^3$. Our main result is an Alexandrov-type uniqueness theorem for this non-convex polyhedron: a deterministic Gröbner-basis computation over $\mathbb{Q}$ shows that the symmetric length equations have exactly four complex solutions, all of them copies of $P$. Homotopy continuation and 6.4 million random starts support the conclusion.

## 1. Introduction

Mizhaev's polyhedron $P$ [26, 25, 35] has $V = 24$ vertices with integer coordinates, $E = 36$ edges and $F = 8$ planar simple nonagons. Three faces meet at every vertex, so the face structure is an equivelar map of type $\{9, 3\}$, and

$$\chi = 24 - 36 + 8 = -4, \qquad g = 3.$$

Every pair of faces is adjacent, twenty pairs along one edge and eight along two. The rotoreflection

$$\mathsf{T}(x, y, z) = (y, -x, -z), \qquad \mathsf{T}^4 = \mathrm{id}, \qquad \det \mathsf{T} = -1, \tag{1}$$

maps the vertex set to itself and generates the full automorphism group $C_4$ of the skeleton [35]. Figure 1 shows the object. At first sight it looks like a rendering error. Our aim is to show that underneath it sits a perfectly calm hyperbolic surface, and to measure how much information is needed to get from one to the other and back.

*Main results.* A *realisation* of the intrinsic geometry of $P$ on its triangulation $K$ (Section 2) is a configuration $X \in (\mathbb{R}^3)^{24}$ with the same 84 edge lengths as $P$; it is a polyhedral surface isometric to $P$, whose nonagons may fold along the diagonals of $K$.

**Theorem A** (Symmetric uniqueness)**.** Among all realisations $X$ of the intrinsic geometry of $P$ on $K$ with $X_{\tau(v)} = \mathsf{T}X_v$ for every vertex $v$, the polyhedron $P$ is the only one, up to congruence. The symmetric length equations have exactly four complex solutions; all of them are real, simple, and copies of $P$.

**Theorem B** (Rigidity)**.** $P$ is infinitesimally rigid, and the edge graph of $K$ is generically globally rigid in $\mathbb{R}^3$.

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Both theorems are computer-assisted and exact. Theorem 11 proves Theorem A with a deterministic Gröbner basis over $\mathbb{Q}$, confirmed independently by msolve and by homotopy continuation; Proposition 8 and Theorem 9 prove Theorem B in exact integer arithmetic. Theorem B does not imply Theorem A, because $P$ is not generic, and no positive semidefinite stress certifies the uniqueness of $P$ (Remark 10). Without symmetry, 6.4 million random starts find no realisation other than $P$.

**Numerical result C** (The canonical twin)**.** The uniformisation $h_P$ of $P$, decorated with the 24 cone points and angles, determines the flat metric of $P$; numerically the round trip closes to $10^{-14}$ on four refinement levels. The circle-packing structure $h_{\mathrm{cp}}$ of $K$ is a different point of Teichmüller space, at Thurston distance at least 0.23 from $h_P$, and the two differ most around the narrow ring of $P$.

Numerical result C is Proposition 6 together with Tables 3 and 4; it is numerical because the uniformisation is computed on finite meshes.

*Related work.* Alexandrov's theorem [1] settles uniqueness for convex polyhedra; for frameworks, generic global rigidity is characterised by Connelly and by Gortler, Healy and Thurston [14, 17], but these results say nothing about a particular non-generic polyhedron such as $P$. Discrete uniformisation [6, 19, 18] and circle packing [4, 11] provide the two routes to a hyperbolic structure that we compare. Mizhaev's coordinates [26, 25] make $P$ an ideal test case: they are integers, so rigidity questions can be decided in exact arithmetic.

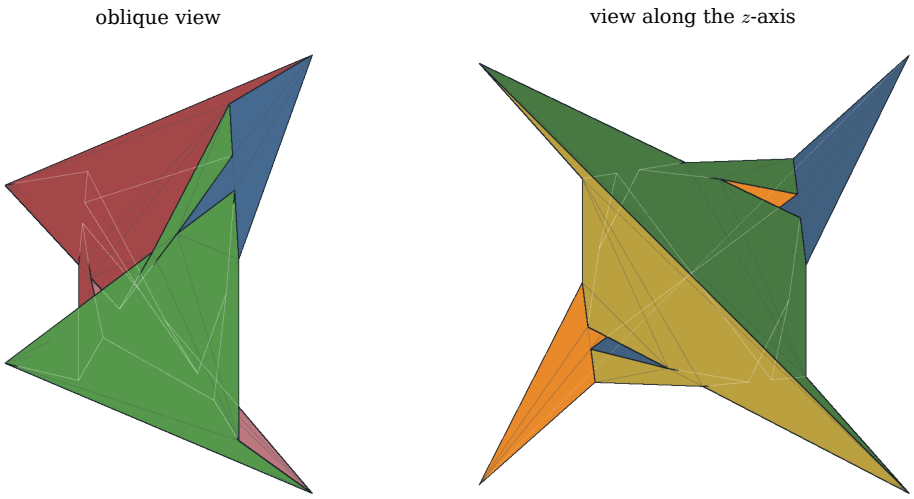



FIGURE 1. Mizhaev's genus-three polyhedron with exact integer coordinates. Colours mark the eight nonagons and match Fig. 4; thin lines are the triangulating diagonals, dashed light lines are hidden nonagon edges. The right panel looks down the axis of $\mathsf{T}$ and shows the four-fold pinwheel symmetry.

*Three handles.* Pictures of $P$ suggest more than three holes. The genus counts independent holes, however, and openings seen from different directions need not be independent. Horizontal sections show what is going on (Fig. 2). From the top down, the solid bounded by $P$ is a single thin wall that splits into two walls. These join again just above $z = 18$ and open into a ring around the axis of $\mathsf{T}$. Below $z = -18$ the ring closes, splits into two walls turned by 90°, and these merge into a single wall at the bottom. The upper and the lower split-and-join each create a handle, and the ring a third one, through which the axis passes; $\mathsf{T}$ exchanges the upper and the lower handle. Section 8 makes this precise with a height function whose Reeb graph has exactly three independent cycles.

Everything below is computed from the coordinates and face walks in [26, Tables 1–2], unchanged. Scripts and full-precision data accompany the paper (Section 12).

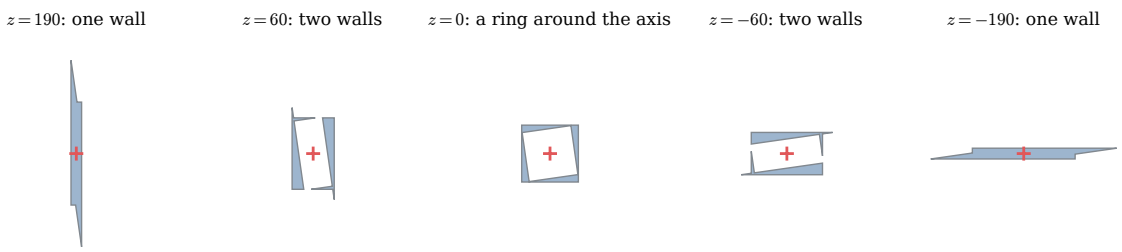



FIGURE 2. Horizontal sections of the solid bounded by $P$ (shaded) at five heights; the cross marks the axis of $\mathsf{T}$. The ring at $z = 0$ is the middle handle. The upper and the lower handle each consist of a wall that splits into two and joins again.

## 2. Triangulating the nonagons

Each nonagon is planar, so it can be split into seven Euclidean triangles by six non-crossing diagonals. We triangulate one representative of each of the two face orbits and transport the result by $\mathsf{T}$. The resulting complex $K$ has

$$(V_K, E_K, F_K) = (24, 84, 56),$$

every edge lies in exactly two triangles, and $\mathsf{T}$ acts on $K$ by a simplicial automorphism $\tau$ of order four.

*Remark* 1 (A cousin of Klein's quartic). The numbers $24, 84, 56$ are forced: any triangulation of a closed genus-three surface with 24 vertices has them, and the average vertex degree is $2 \cdot 84/24 = 7$. The most symmetric such triangulation is the $\{3, 7\}$ tiling of Klein's quartic [21], with every degree equal to 7. Our $K$ has the same counts but degrees 6 (sixteen vertices), 7 (four) and 11 (four). It is a lopsided cousin of Klein's surface, and the circle radii below measure how lopsided.

**Lemma 2** (The symmetry, intrinsically)**.** *$\tau$ and $\tau^3$ act on $|K|$ without fixed points; $\tau^2$ fixes exactly the midpoints of the four edges $\{5, 8\}$, $\{18, 22\}$, $\{19, 23\}$, $\{20, 24\}$, each of which it reverses. Consequently $|K|/\langle\tau^2\rangle$ is a torus, and $|K|/\langle\tau\rangle$ is a Klein bottle with two cone points of order two, of orbifold Euler characteristic $-1 = \chi(K)/4$.*

*Proof.* A direct check of the permutation shows that no vertex, edge or triangle is mapped to itself by $\tau$ or $\tau^3$, and that $\tau^2$ maps exactly the four listed edges to themselves with reversed endpoints. A simplicial map fixing no simplex setwise has no fixed point, which gives the first claim. Riemann–Hurwitz for the orientation-preserving involution $\tau^2$ with four fixed points gives $-4 = 2\chi' - 4$, so $\chi' = 0$, a torus. The induced involution on that torus reverses orientation (since $\det \mathsf{T} = -1$) and has no fixed points (a fixed point would be fixed by $\tau$), so the quotient is a Klein bottle; the four fixed points of $\tau^2$ form two $\tau$-orbits and become two cone points of angle $\pi$. As a check, the Lefschetz numbers $L(\tau) = 1 - 0 + (-1) = 0$ and $L(\tau^2) = 1 - (-2) + 1 = 4$, computed from the homology traces in Section 11, agree with the fixed-point counts. □

## 3. The circle-packing structure

Now forget the Euclidean shape and keep only the combinatorics of $K$. Give each vertex $i$ a radius $r_i > 0$ and declare the edge $\{i, j\}$ to be a hyperbolic geodesic of length $\ell_{ij} = r_i + r_j$. Each triangle $(i, j, k)$ becomes a hyperbolic triangle whose angle at $i$ follows from the hyperbolic law of cosines,

$$\cos \alpha_i = \frac{\cosh \ell_{ij} \cosh \ell_{ik} - \cosh \ell_{jk}}{\sinh \ell_{ij} \sinh \ell_{ik}}. \tag{2}$$

Circles of radius $r_i$ about the vertices are then mutually tangent along every edge. We ask that every vertex be flat:

$$\Theta_i(r) := \sum_{(i,j,k)\in K} \alpha_i(r) = 2\pi \qquad (i = 1, \dots, 24). \tag{3}$$

By the discrete uniformisation theorem for circle packings [31, 4, 11], for a triangulation of a closed surface of genus $g \geq 2$ this system has exactly one solution, and the glued triangles form a closed hyperbolic surface $(S, h_{\mathrm{cp}})$ whose metric is smooth across the vertices because there is no angle deficit left.

Equation (3) is 24 equations in 24 unknowns. Starting from $r_i = 0.4$, a trust-region least-squares solve in $u_i = \log r_i$ returns

$$\max_i |\Theta_i - 2\pi| = 3.6 \cdot 10^{-15}.$$

Floating point is not proof, so we certify the solution.

**Theorem 3** (computer-assisted)**.** *The system* (3) *has exactly one solution in a box of radius* $10^{-60}$ *around the computed* $\log r$*. All* 24 *radii are enclosed in balls of radius* $1.1 \cdot 10^{-60}$*; the values in Table 1 are correct in every printed digit.*

*Proof.* Let $F(u) = \Theta(e^u) - 2\pi$ and let $X$ be the box. The Krawczyk operator $K(X) = \tilde{u} - YF(\tilde{u}) + (I - YF'(X))(X - \tilde{u})$, with $\tilde{u}$ the midpoint and $Y$ any fixed matrix, satisfies $K(X) \subset \operatorname{int} X$ when evaluated with rigorous enclosures; by Krawczyk's theorem [22] $F$ then has exactly one zero in $X$. We evaluate $F$ and its analytic Jacobian in Arb ball arithmetic [20] at 256 and 512 bits; the inclusion holds at both precisions and was reproduced on two machines. □

The solution has only six distinct radii (Table 1). This is not a numerical accident: $\tau$ maps one solution of (3) to another, so uniqueness forces $r_{\tau(i)} = r_i$, and $\tau$ becomes an orientation-reversing isometry of $(S, h_{\mathrm{cp}})$. Large circles go where many triangles must share 360°, because in the hyperbolic plane bigger triangles have smaller angles. For comparison, Klein's all-degree-seven triangulation packs with equal radii $r_{\mathrm{K}} = \frac{1}{2}\operatorname{arccosh}\big(\cos\frac{2\pi}{7}/(1-\cos\frac{2\pi}{7})\big) = 0.5453$; our degree-seven vertices come close.

TABLE 1. The six vertex orbits of $\tau$. The Euclidean angle sum is that of the polyhedron $P$; the hyperbolic one is 360° for every vertex.

| Orbit (Mizhaev's labels) | degree in $K$ | angle sum on $P$ | hyperbolic radius $r_i$ |
|---|---|---|---|
| $5, 8, 20, 24$ | 6 | 95.53° | 0.3827183071 |
| $18, 19, 22, 23$ | 6 | 457.68° | 0.4303128653 |
| $1, 3, 14, 21$ | 6 | 408.13° | 0.4403091423 |
| $2, 4, 10, 15$ | 6 | 479.77° | 0.4540050625 |
| $7, 12, 13, 17$ | 7 | 396.34° | 0.5144900382 |
| $6, 9, 11, 16$ | 11 | 682.55° | 0.8802882148 |

Figure 3 shows what happens at the two extreme vertices. On the polyhedron, the six triangles at vertex 5 fill only 95.5°, a sharp spike. The eleven triangles at vertex 6 fill 682.6° and wind almost twice around the point. In the hyperbolic twin both flowers close up to exactly one full turn.

*Remark* 4 (An honest check). The hyperbolic area $\sum_\Delta(\pi - \alpha - \beta - \gamma) = 56\pi - \sum_i \Theta_i$ equals $8\pi = 4\pi(g-1)$ as soon as (3) holds. Our computed area 25.13274122871834 is therefore a consequence of the small residual, not an independent test of it.

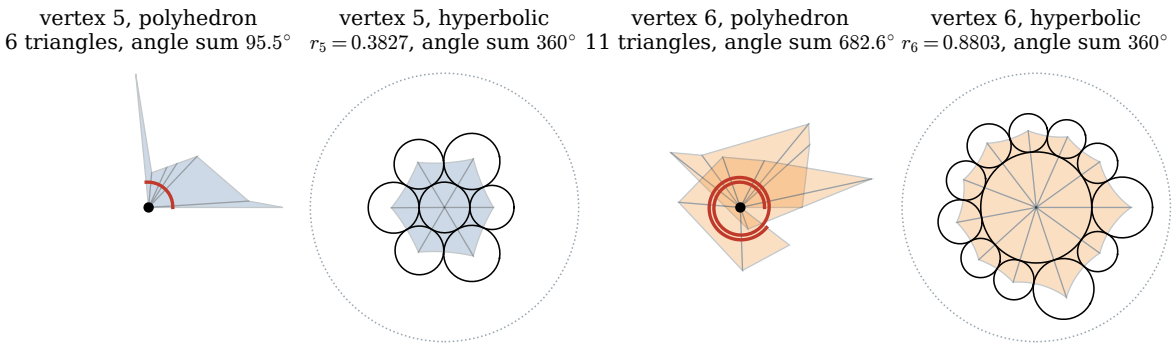



FIGURE 3. Before and after at a spike (vertex 5) and at a saddle (vertex 6). Left of each pair: the Euclidean triangles of $K$ unfolded around the vertex, with the red arc showing the total angle. Right of each pair: the same triangles in the Poincaré disk with their tangent circles; the angle sum is $2\pi$ up to $4 \cdot 10^{-15}$.

## 4. Development in the Poincaré disk

The surface $(S, h_{\mathrm{cp}})$ is intrinsic; it is not a shape in space. To see it, we develop it into the Poincaré disk. Choose a spanning tree of the dual graph of $K$, place one triangle, and attach the others one at a time across tree edges using (2) and disk automorphisms $z \mapsto e^{i\theta}(z+a)/(1+\bar{a}z)$. We use a tree that keeps each nonagon in one piece. The result (Fig. 4) is a fundamental domain made of 56 geodesic triangles, with the $84 - 55 = 29$ non-tree edges appearing twice on its boundary, to be glued in pairs.

Figure 4 is the smooth surface the title promises. Its curvature is $-1$ everywhere, including at the former spikes and saddles, and its area is $8\pi$. It carries an order-four isometry with the orbifold quotient of Lemma 2, a hyperbolic Klein bottle with two cone points of area $2\pi$.

*Remark* 5 (What the twin forgets). The radii were computed from $K$ alone; the integer coordinates never entered (3). Any other polyhedral realisation of the same triangulation, spiky or not, has the same hyperbolic twin. Going there is free; coming back costs data.

## 5. The uniformisation of $P$

The circle-packing twin has a blemish: it is not canonical. Its radii depend on $K$, and $K$ depends on how each nonagon was cut into triangles, a choice made by an ear-clipping routine and by nothing in the geometry. A second hyperbolic surface depends on $P$ alone. The flat metric of $P$, with its 24 cone points, defines a conformal structure on $S$, and by uniformisation this structure carries exactly one metric of curvature $-1$; we call it $h_P$.

We compute $h_P$ discretely. On the $\ell$-th refinement of $K$ (every triangle of $P$ cut into $4^\ell$ similar triangles) we look for vertex values $u$ such that the hyperbolic lengths defined by

$$2 \sinh \tfrac{L_{ij}}{2} = e^{(u_i+u_j)/2} \, |x_i - x_j| \tag{4}$$

make every vertex flat. This is discrete conformal equivalence in the sense of Bobenko, Pinkall and Springborn [6]. Damped Newton converges on levels 0 to 3 (up to 1,788 vertices) to angle residuals below $2.4 \cdot 10^{-10}$, without violating a triangle inequality; the edge flips that the general theory requires [19] were not needed here, and the length comparisons below change by less than 0.5% from level 2 to level 3, in line with known convergence results [18].

To compare the two twins we measure closed geodesics. Both metrics live on the same triangulated surface, so a closed chain of triangles names the same free homotopy class in both. Developing the chain into the hyperboloid model yields a holonomy $A \in SO^+(2,1)$, and the length of the closed geodesic follows from $\operatorname{tr} A = 1 + 2\cosh \ell$. As a check of the tool, the lengths in $h_{\mathrm{cp}}$ computed on levels 0, 1 and 2 agree to $10^{-6}$. Twenty non-trivial chains and all reduced words of length at most three in them give 62,306 test curves.

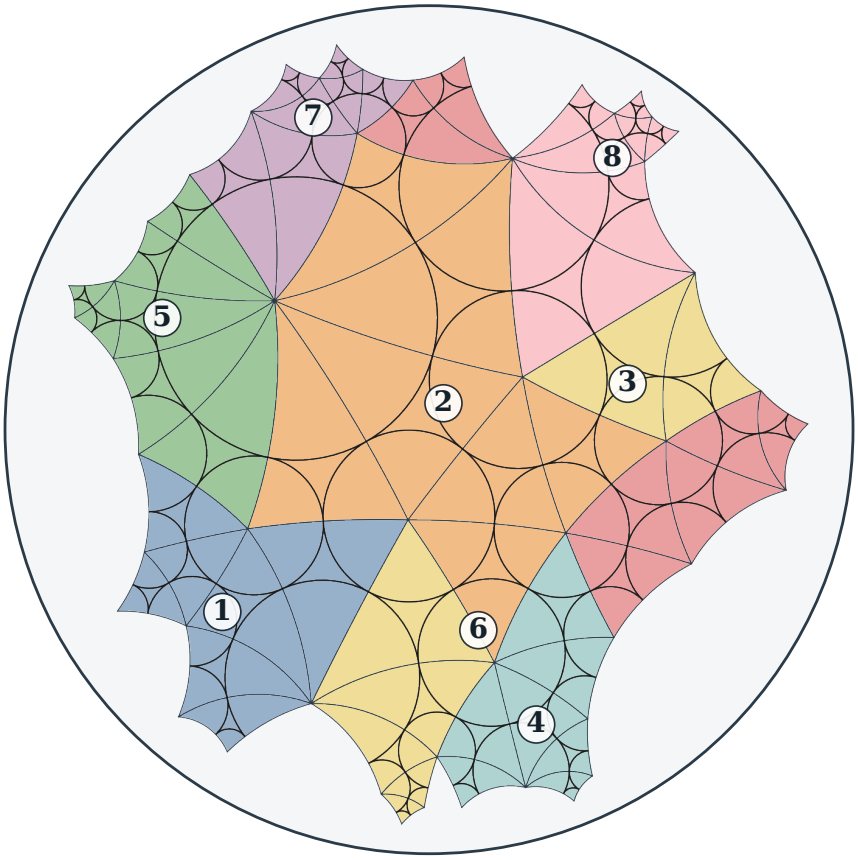



FIGURE 4. The hyperbolic twin of Mizhaev's polyhedron, unfolded in the Poincaré disk. All 56 triangles are geodesic, all circles are tangent along every edge, and the angles around each vertex add up to $2\pi$. Numbers identify the eight nonagons, coloured as in Fig. 1. The central nonagon touches all seven others: Mizhaev's "every two faces are adjacent" becomes visible in a single picture.

TABLE 2. The four shortest test geodesics of $h_{\mathrm{cp}}$ and the same homotopy classes in $h_P$, computed on refinement levels 0 to 3.

| $h_{\mathrm{cp}}$ | $h_P$, level 0 | level 1 | level 2 | level 3 |
|---|---|---|---|---|
| 2.984 | 3.256 | 3.238 | 3.197 | 3.188 |
| 3.384 | 3.044 | 3.082 | 3.099 | 3.101 |
| 4.330 | 5.336 | 5.242 | 5.181 | 5.182 |
| 4.433 | 5.703 | 5.638 | 5.593 | 5.580 |

**Proposition 6** (numerical)**.** *Over all test curves the ratio* $\ell_{h_P}/\ell_{h_{\mathrm{cp}}}$ *lies in* $[0.7964, 1.2592]$ *at level* 3*; the bound* $\log\max$ *takes the values* 0.252, 0.240, 0.232, 0.230 *on levels* 0 *to* 3*. Hence Thurston's asymmetric distance* [32]*,* $d(X,Y) = \log\sup_\gamma \ell_Y(\gamma)/\ell_X(\gamma)$*, satisfies*

$$d(h_{\mathrm{cp}}, h_P) \geq 0.230, \qquad d(h_P, h_{\mathrm{cp}}) \geq 0.228,$$

*up to the discretisation error of* $h_P$*, which the level sequence puts at the third decimal.*

The two twins are therefore different points of the 12-dimensional Teichmüller space of $S$, and the systoles differ too (2.984 against 3.086). Figure 6 shows the whole comparison. A finite set of curves can only bound the distance from below; the true distance is at least as large.

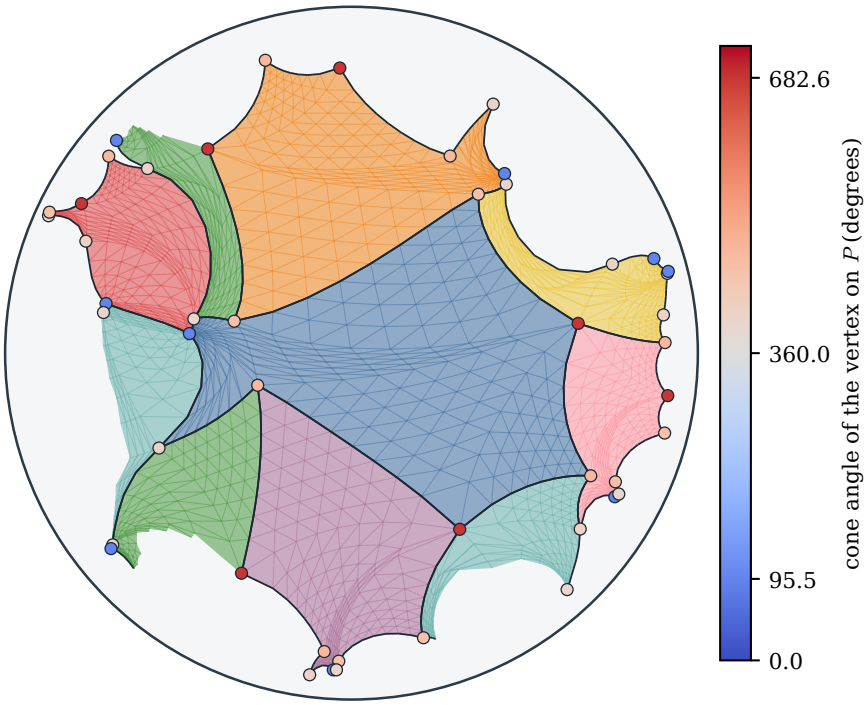



FIGURE 5. The canonical twin $h_P$ (refinement level 3, 1,788 vertices) developed in the Poincaré disk. Colours: the nonagons, as in Fig. 1; a nonagon may appear in two pieces because the surface is cut open. Dark curves: the edges of $P$, which are not geodesics of $h_P$. Dots: the 24 cone points of $P$, coloured by their cone angle; on $h_P$ every one of them has total angle $2\pi$.

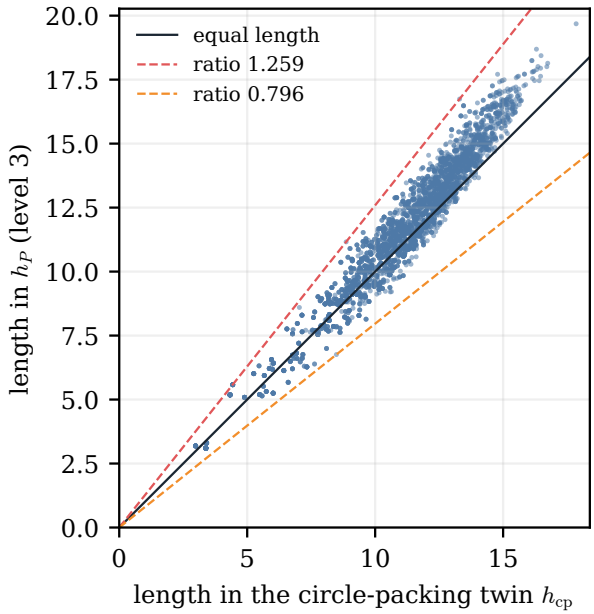



FIGURE 6. Lengths of 3,933 distinct test geodesics in the two twins. Dashed lines bound the ratio; the points would lie on the solid diagonal if the twins were isometric.

## 6. Inverting the uniformisation

The forward map $P \mapsto h_P$ forgets where the corners of $P$ were and how sharp they were. Keeping exactly that information makes the map invertible. Let $p_1, \dots, p_{24} \in S$ be the cone points and $\theta_i$ their cone angles; on $P$ they satisfy $\sum_i (2\pi - \theta_i) = 2\pi\chi(S) = -8\pi$.

**Theorem 7** (Troyanov [33])**.** *On a closed Riemann surface with points $p_i$ and angles $\theta_i > 0$ satisfying the Gauss–Bonnet condition there is exactly one flat metric with cone singularities of angle $\theta_i$ at $p_i$ in the given conformal class, up to scaling.*

Together with uniformisation this gives a canonical correspondence between the intrinsic geometry of $P$ and the decorated hyperbolic surface,

$$\text{flat cone metric of } P \;/\; \text{scale} \quad \longleftrightarrow \quad \big(h_P;\ p_1, \dots, p_{24};\ \theta_1, \dots, \theta_{24}\big).$$

The parameter count matches: flat metrics on the triangulation $K$ have 84 edge lengths, 83 up to scale, and the right-hand side has 12 (Teichmüller space) $+\,48$ (point positions) $+\,24$ (angles) $-\,1$ (Gauss–Bonnet) $= 83$ parameters.

We carry out the inverse discretely. Given only the $\lambda$-lengths of $h_P$ on the $\ell$-th refinement, the 24 marked vertices and their angles, we solve for $u$ such that the Euclidean lengths $e^{(u_i+u_j)/2}\, 2\sinh(L_{ij}/2)$ have angle sum $\theta_i$ at the marked vertices and $2\pi$ elsewhere, using the continuous extension of the angle function to degenerate triangles [6]. The problem is convex, so its solution is unique up to scale. Table 3 compares the result with the edge lengths of $P$.

Table 3. Round trip $P \to (h_P, p, \theta) \to P$ on four refinement levels. Residuals are maximal angle defects; the last column is the largest relative deviation of the recovered edge lengths from those of $P$ after fitting the one free scale.

| level | vertices | forward residual | inverse residual | edge-length error |
|---|---|---|---|---|
| 0 | 24 | $2.3 \cdot 10^{-13}$ | $1.1 \cdot 10^{-14}$ | $2.0 \cdot 10^{-15}$ |
| 1 | 108 | $1.8 \cdot 10^{-13}$ | $9.9 \cdot 10^{-14}$ | $4.2 \cdot 10^{-15}$ |
| 2 | 444 | $2.4 \cdot 10^{-12}$ | $7.9 \cdot 10^{-13}$ | $5.4 \cdot 10^{-14}$ |
| 3 | 1,788 | $2.4 \cdot 10^{-10}$ | $2.2 \cdot 10^{-14}$ | $9.9 \cdot 10^{-15}$ |

*What kind of map is this?* The correspondence is a bijection between two 83-dimensional spaces, not a homomorphism: neither side carries a group structure. What it respects are symmetries. An isometry of $P$ becomes an isometry of $h_P$ that permutes the marked points, conformal or anticonformal according to whether it preserves orientation, and this assignment is an injective group homomorphism from the symmetry group of $P$ to the isometry group of $(h_P; p_1, \dots, p_{24})$. For the rotoreflection $\mathsf{T}$ the image acts on $H_1(S;\mathbb{R})$ with character $(6, 0, -2, 0)$ on $\tau^0, \dots, \tau^3$.

*What is new and what is not.* Uniformisation and Troyanov's theorem are classical, and so is the correspondence above. Its combinatorial counterpart, between ribbon graphs with lengths on their edges and hyperbolic surfaces with cusps, goes back to Penner [27] and to Bowditch and Epstein [7]. The ribbon graph of $P$ alone (Section 8) would not suffice: it records only which faces meet where, and polyhedra with the same ribbon graph but other coordinates have different twins. What this paper adds is the explicit computation for one concrete, strongly non-convex example, with certificates and convergence checks; the measured distance between the canonical twin and the circle-packing twin; and, in Section 9, the question whether the last step back into space is unique.

At the discrete level this exactness is guaranteed by convexity; its value lies in showing, concretely and on several levels, that the decorated hyperbolic surface carries the intrinsic geometry of $P$, no more and no less. One step is left: from the flat metric back to a polyhedron in $\mathbb{R}^3$.

## 7. Surfaces in space conformal to $h_P$

The correspondence of Section 6 is intrinsic, and it is worth saying what that means for pictures. Every surface in $\mathbb{R}^3$ whose conformal structure is that of $P$ is a view of the same object. Uniformisation sends each of them to $h_P$, and the 24 marked points travel with the vertices of $P$. The polyhedron itself is one such view, with all of its curvature concentrated at the cone points. Figure 7 shows two more: embedded, four-fold symmetric surfaces whose edge lengths are discretely conformal to those of $P$, and which differ only in how much bending they are allowed. One stays close to $P$ and keeps its creases; the other is rounder and moves further away. Shape, distance to $P$ and bending energy are properties of the view, not of the object. No view is singled out either, because conformal embeddings of a Riemann surface come in infinite-dimensional families [16]. The surfaces of the circle-packing searches, by contrast, are views of a different object, $h_{\mathrm{cp}}$, which Proposition 6 places at Thurston distance at least 0.23.

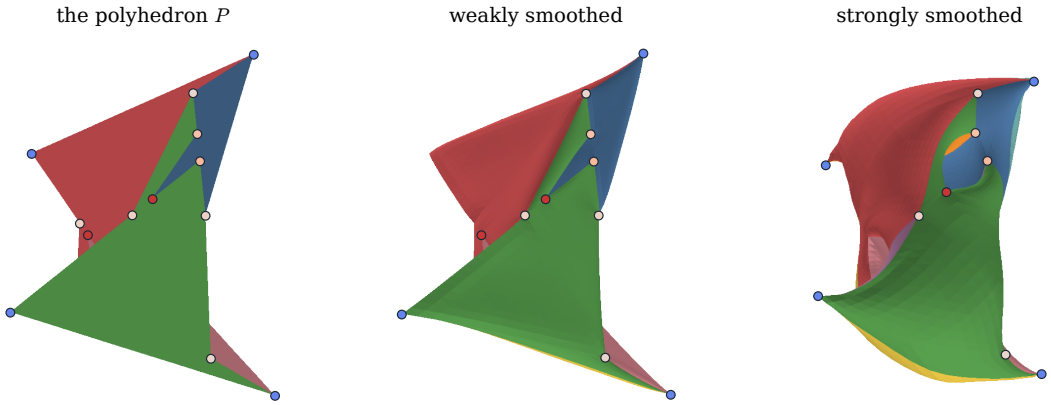



Figure 7. Three views of one object. Left: the polyhedron $P$. Middle and right: embedded, exactly four-fold symmetric surfaces whose edge lengths are discretely conformal to those of $P$, with weak and strong smoothing. All three have the same uniformisation $h_P$, with the same 24 marked points (dots, coloured by the cone angle on $P$ as in Fig. 5); on a smooth view these are ordinary points. Refinement level 4 (7,164 vertices); conformal residual $1.8 \cdot 10^{-4}$ and $5.1 \cdot 10^{-3}$, mean distance to $P$ 4.1 and 27 units (maximal 17 and 88), Willmore energy $25 \cdot 4\pi$ and $8.2 \cdot 4\pi$; both checked independently for embeddedness and symmetry.

## 8. Pants decomposition and ribbon graph

A closed hyperbolic surface of genus 3 can be cut along $3g - 3 = 6$ disjoint simple closed geodesics into $2g - 2 = 4$ pairs of pants, spheres with three holes. By Gauss–Bonnet every pair of pants has area $2\pi$, so the four always have the same area, whatever the surface. Their shapes are fixed by the lengths of their three cuffs, and six lengths together with six twist parameters form the Fenchel–Nielsen coordinates of Teichmüller space [10, Ch. 3 and 6].

We read a decomposition off the shape of $P$. The height $z + \varepsilon x + \varepsilon' y$, with small $\varepsilon, \varepsilon'$ that separate the vertex heights, is a piecewise linear Morse function on $P$ in the sense of Banchoff [3]; the indices of its critical vertices add up to $\chi = -4$. In its Reeb graph we remove the branches leading to maxima and minima, which bound discs, and suppress vertices of degree two. What remains is a trivalent graph with four vertices and six edges, the dual graph of a pants decomposition, with each cuff represented by a level curve on $P$. Its shape matches the three handles of Fig. 2: at the bottom and at the top a pair of pants has two of its cuffs glued to each other, and two pants in the middle are joined along the inner and the outer rim of the ring. Table 4 lists the lengths of the closed geodesics in the six cuff classes, computed from holonomies as in Section 5.

TABLE 4. The six cuffs of the pants decomposition: lengths of the closed geodesics in their free homotopy classes, in the circle-packing twin and in the canonical twin on three refinement levels. Last column: stretch factor from $h_{\mathrm{cp}}$ to $h_P$.

| cuff | level curve | $h_{\mathrm{cp}}$ | $h_P$, level 1 | level 2 | level 3 | ratio |
|---|---|---|---|---|---|---|
| $\gamma_1$ | lower handle, $z \approx -60$ | 2.984 | 3.238 | 3.197 | 3.188 | 1.068 |
| $\gamma_2$ | below the ring, $z \approx -18$ | 9.298 | 9.970 | 9.880 | 9.865 | 1.061 |
| $\gamma_3$ | inner rim of the ring, $z = 0$ | 4.433 | 5.638 | 5.593 | 5.580 | 1.259 |
| $\gamma_4$ | outer rim of the ring, $z = 0$ | 5.972 | 6.004 | 5.915 | 5.907 | 0.989 |
| $\gamma_5$ | above the ring, $z \approx 18$ | 9.298 | 9.970 | 9.880 | 9.865 | 1.061 |
| $\gamma_6$ | upper handle, $z \approx 102$ | 2.984 | 3.238 | 3.197 | 3.188 | 1.068 |

Since $\mathsf{T}$ maps $z$ to $-z$, the symmetry exchanges the lower and the upper half of the decomposition: $\gamma_1, \gamma_6$ and $\gamma_2, \gamma_5$ have equal lengths, and the images under $\tau$ of all six cuffs have the same lengths as the cuffs themselves. There are therefore two congruence classes of pants, the handle pants with cuffs $(\gamma_1, \gamma_1, \gamma_2)$ and the middle pants with cuffs $(\gamma_2, \gamma_3, \gamma_4)$, and all four have area $2\pi$. The decomposition depends on the choice of height function; another choice gives other shapes, but always the same areas. We have not computed the twist parameters, so the table gives half of the Fenchel–Nielsen coordinates.

The last column is the most informative. The inner rim $\gamma_3$ of the ring is stretched by the factor 1.259 from $h_{\mathrm{cp}}$ to $h_P$, which is, to three decimals, the largest ratio among all 62,306 test curves of Proposition 6. The two twins differ most around the hole through which the symmetry axis passes, where the circle packing, which sees only combinatorics, cannot know how narrow the ring of $P$ is.

*The ribbon graph.* The edges of $P$ form a trivalent graph, and the orientation of the surface gives a cyclic order of the three edges at each vertex. This ribbon graph, listed in Section A, records the combinatorics of $P$ completely: its eight boundary cycles are exactly the nonagons, and $24 - 36 + 8 = -4$. The symmetry maps it to itself and reverses the cyclic order at all 24 vertices, as an orientation-reversing map must. It does not determine $h_P$, which also depends on the coordinates; with lengths on its edges it would correspond to a hyperbolic surface with eight cusps [27, 7], and without lengths the construction of Brooks and Makover [9] turns it into one particular closed surface (Section 11). The dark curves in Fig. 5 are this graph drawn in $h_P$; Fig. 8 shows it as a fat graph in the plane.

## 9. UNIQUENESS OF THE REALISATION

Cauchy's theorem and Alexandrov's uniqueness theorem [1, Ch. 3] say that a convex polyhedron is determined, up to congruence, by its intrinsic metric. For non-convex polyhedra this can fail: pushing in a vertex of a convex polyhedron gives an isometric but non-congruent one, and Connelly's flexible polyhedron [12] even moves through a continuous family of isometric shapes. Neither situation is excluded for $P$ a priori. We ask whether another polyhedral surface in $\mathbb{R}^3$ is built from the same 56 triangles of $K$, glued in the same way, so that $|X_i - X_j|^2 = D_{ij}$ for all 84 edges; we call such an $X$ a *realisation*. The $D_{ij}$ are integers, the nonagons may fold along their diagonals, and every realisation is isometric to $P$.

**Proposition 8** (Local uniqueness)**.** *$P$ is infinitesimally rigid: the rigidity matrix of the 84 length conditions has rank $66 = 3 \cdot 24 - 6$ at $P$. Consequently every realisation sufficiently close to $P$ is congruent to $P$.*

*Proof.* $P$ has integer coordinates, so its rigidity matrix has integer entries and its rank can be computed exactly; it is 66. Numerically the conclusion is robust as well: the smallest non-zero singular value is 0.065 for unit edge directions. Infinitesimal rigidity implies local rigidity [2]. □

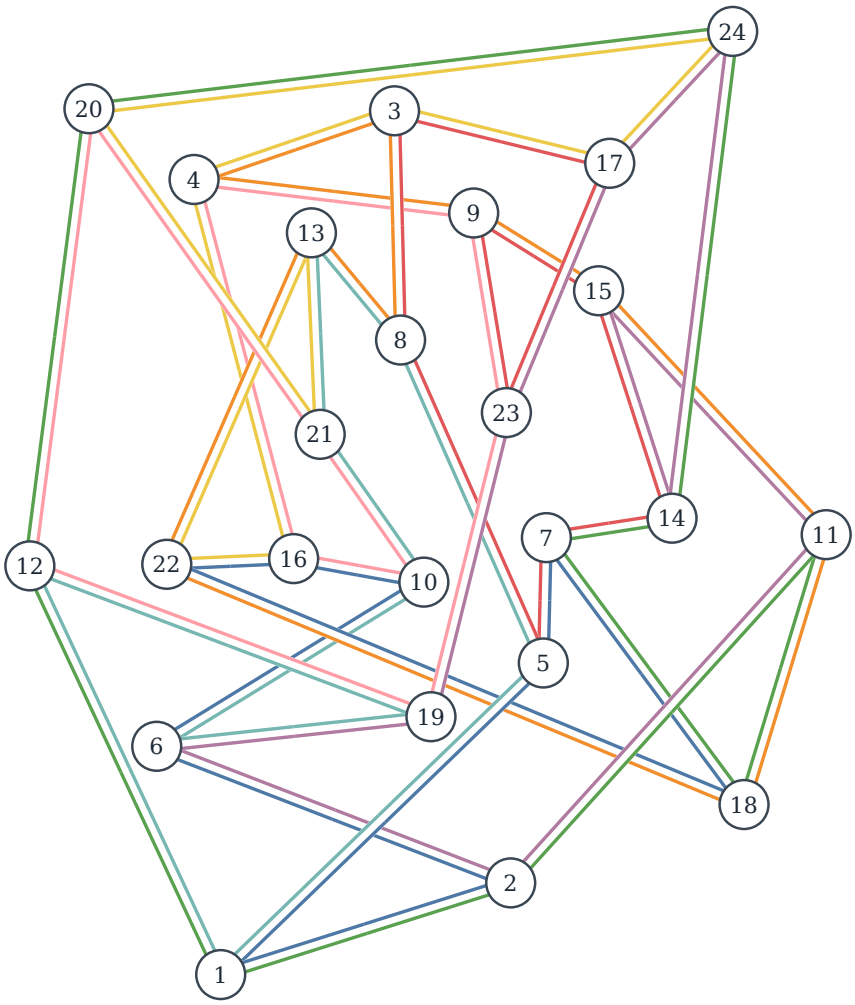



FIGURE 8. The ribbon graph of $P$ as a fat graph: the 36 edges drawn as bands between the 24 vertices, with the cyclic order at every vertex as on the surface. The two rims of each band carry the colours of the two nonagons that meet along that edge, as in Fig. 1; following one colour along the rims traces one nonagon. A ribbon graph of genus 3 cannot be drawn in the plane without crossings or twisted bands, since a drawing with neither would make it planar. This drawing has 14 crossings, shown as over- and underpasses, and no twisted band. Vertex numbers as in [26].

**Theorem 9** (Generic global rigidity)**.** *The edge graph of $K$ is generically globally rigid in $\mathbb{R}^3$: for a generic configuration $p$, every configuration with the same* 84 *edge lengths is congruent to $p$.*

*Proof.* In exact integer arithmetic at $P$, the equilibrium stresses, the left kernel of the rigidity matrix, form an 18-dimensional space with an integer basis, and a random integer stress $\omega_0$ has a stress matrix $\Omega(\omega_0)$ of rank $20 = n - d - 1$; five independent choices all give rank 20. Since the rigidity matrix has rank 66 on a neighbourhood $U$ of $P$, the stress spaces form a vector bundle of rank 18 over $U$, and $\omega_0$ extends to a continuous family of stresses $\omega(p)$. By lower semicontinuity of the rank, $\operatorname{rank}\Omega(\omega(p)) \geq 20$ near $P$, and the rank is at most $n - d - 1 = 20$ for every configuration that spans $\mathbb{R}^3$, since the kernel of a stress matrix contains the constant vector and the three coordinate vectors. A generic $p$ near $P$ therefore carries a stress of rank $n-d-1$, so $(G, p)$ is globally rigid by Connelly's theorem [14]. By Gortler, Healy and Thurston [17], either all generic frameworks of a graph are globally rigid or none is. □

*Remark* 10 (Why this does not settle $P$). $P$ is far from generic: it has integer coordinates, an order-four symmetry and 64 collinear vertex triples inside its nonagons. Connelly's sufficient condition for the global rigidity of a particular framework [13, 14] asks for a positive semidefinite stress of rank 20 and for edge directions that do not lie on a conic at infinity. The second condition holds at $P$. The first fails: a semidefinite program over the stress space of $P$ shows that the largest attainable smallest eigenvalue on the relevant 20-dimensional complement is zero, $5 \cdot 10^{-11}$ with one solver and $-5 \cdot 10^{-9}$ with another. Uniqueness of $P$ therefore needs a different argument.

*The symmetric problem, numerically.* Requiring the symmetry $X_{\tau(v)} = \mathsf{T}X_v$ leaves six orbit representatives, and fixing the rotation about the axis leaves 17 unknowns, constrained by 22 quadratic equations, one per edge orbit. We randomised the system to 17 equations and solved it by projective total-degree homotopy continuation [28] over all $2^{17} = 131{,}072$ start points. This method reaches every isolated nonsingular complex solution with probability one. The tracker was validated on dense random quadratic systems, where it finds all $2^n$ solutions. Five runs with independent random parameters each lose 3 to 5% of their paths, but the union of their solution sets reaches 22,112 distinct finite solutions after three runs and does not grow in the fourth and fifth. Among them, exactly four satisfy all 22 original equations, all four are real, and all four are copies of $P$ under the rotation by $\pi$ about the axis and the reflection $z \mapsto -z$, which commute with $\mathsf{T}$.

*The symmetric problem, algebraically.* Written in unscaled coordinates, the 22 symmetric equations have integer coefficients, so they can be treated by Gröbner bases over $\mathbb{Q}$. The four copies of $P$ are explicit solutions. The rotation about the axis that takes the first orbit representative, vertex $1 = (-72, 84, 18)$, into the $xz$-plane is

$$(x, y, z) \longmapsto \left( \tfrac{7y-6x}{\sqrt{85}},\ -\tfrac{7x+6y}{\sqrt{85}},\ z \right),$$

since $72^2 + 84^2 = 144 \cdot 85$. It maps the six orbit representatives to the points in Table 5, all with coordinates in $\mathbb{Q}(\sqrt{85})$; the other three solutions follow by the rotation by $\pi$ about the axis and the reflection $z \mapsto -z$, which commute with $\mathsf{T}$ and preserve the normalisation. Substituted into the 22 equations, these values give zero exactly.

TABLE 5. The solution of the symmetric system in normal form: the six orbit representatives of $P$ (Mizhaev's labels), rotated so that vertex 1 lies in the $xz$-plane. The remaining 18 vertices follow from $X_{\tau(v)} = \mathsf{T}X_v$.

| vertex | $x$ | $y$ | $z$ |
|---|---|---|---|
| 1 | $12\sqrt{85}$ | $0$ | $18$ |
| 2 | $\frac{200}{17}\sqrt{85}$ | $-\frac{84}{17}\sqrt{85}$ | $102$ |
| 5 | $\frac{420}{17}\sqrt{85}$ | $-\frac{360}{17}\sqrt{85}$ | $234$ |
| 6 | $\frac{168}{17}\sqrt{85}$ | $\frac{60}{17}\sqrt{85}$ | $-18$ |
| 7 | $\frac{195}{17}\sqrt{85}$ | $-\frac{189}{17}\sqrt{85}$ | $207$ |
| 18 | $\frac{198}{17}\sqrt{85}$ | $-\frac{126}{17}\sqrt{85}$ | $144$ |

**Theorem 11** (Symmetric uniqueness)**.** *Among realisations $X$ on $K$ with $X_{\tau(v)} = \mathsf{T}X_v$ for all vertices $v$, the polyhedron $P$ is the only one, up to congruence. More precisely, the* 22 *symmetric length equations have exactly four complex solutions; all of them are real, simple, and copies of $P$. In particular there is no flexible symmetric realisation.*

*Proof.* The reduced Gröbner basis of the ideal generated by the 22 equations, computed in deterministic exact rational arithmetic with the algorithm slimgb [8] of Singular [15], has 17 elements; the ideal is zero-dimensional and its quotient ring has dimension 4 over $\mathbb{Q}$, so the system has exactly four complex solutions counted with multiplicity. The four copies of $P$ are four distinct solutions, hence they are all solutions, and each is simple. A zero-dimensional ideal admits no continuous family of solutions. □

The computation took 19 minutes on one core; the Gröbner basis is included in the repository. Two further computations agree with it: msolve [5], which works multi-modularly and finds the same zero-dimensional ideal of degree four in 34 seconds, isolating four real solutions that map back to the four copies of $P$ to $2 \cdot 10^{-14}$; and the homotopy continuation described above.

*Without symmetry.* We solved the length equations by Levenberg–Marquardt from 6.4 million starts: uniformly random configurations, $P$ with large noise, and $P$ with vertices reflected through local or global planes. Half of the starts used $K$; the other half used 64 random alternative triangulations of the nonagons, so that the nonagons fold along other diagonals. Of the 977,914 starts that converged to exact solutions, every single one is congruent to $P$.

*Scope.* All statements concern realisations on $K$, surfaces whose creases lie along edges of $K$. A realisation of the intrinsic metric could in principle crease along other geodesics of the flat metric; such realisations, like non-symmetric ones, are covered by the random search only.

## 10. Smooth embeddings and spectral coordinates

*A smooth surface in space.* No smooth closed surface in $\mathbb{R}^3$ is isometric to a hyperbolic surface, since it has a point of positive curvature farthest from the origin. Isometric $C^1$ embeddings exist by Nash–Kuiper [23] but are not smooth; conformal embeddings exist [16], and we searched for symmetric ones on refinements of up to 7,164 vertices, using an incremental-potential contact barrier [24] to keep them embedded. Embedded surfaces discretely conformal to $h_{\mathrm{cp}}$ exist with residual $10^{-6}$, but none comes closer than about 50 units to $P$, which Proposition 6 explains. Conformal to $h_P$, embedded surfaces stay within 4 units of $P$, but then their Willmore energy grows under refinement, from $4.2 \cdot 4\pi$ to $25 \cdot 4\pi$ over four levels: a surface that close to $P$ must keep its creases. Smoothness and closeness trade off against each other, and no particular smooth surface is singled out; Section 7 explains why none needs to be.

*A spectral shortcut.* Expanding the coordinates of $P$ in eigenfunctions of a graph Laplacian on the circle-packing twin [34] is exact but trivial, being a change of basis. Its cost is set by the symmetry: the vertex functions decompose as six copies of $\mathbf{1} \oplus \varepsilon \oplus \rho$, the coordinates of $P$ have no component on the trivial modes, $z$ lives on $\varepsilon$ and $(x, y)$ on $\rho$, so $P$ is encoded by exactly $6 \cdot 1 + 6 \cdot 2 = 18$ real numbers, the same as six vertex orbits times three coordinates.

## 11. Open questions

*Global uniqueness without symmetry.* Theorem 11 and the random search make it very likely that $P$ is the only polyhedron with its intrinsic geometry, but we have no proof. A complete solve without symmetry has 66 unknowns and is out of reach of total-degree homotopy. A positive answer would be a rare uniqueness theorem for a non-convex polyhedron of higher genus; a negative one would show that Mizhaev's object is not determined by its faces and how they are glued.

*Other fold patterns.* The classification uses the diagonals of $K$. Symmetric alternatives with other diagonals inside the nonagons lead to systems of the same kind, which can be solved the same way, one exact solve per pattern.

*Twists.* The Fenchel–Nielsen twists of the decomposition in Section 8 would complete the coordinates of both twins and give a second, independent measure of their distance.

*Certificates.* Theorems 3, 9 and 11 and Proposition 8 are rigorous, computer-assisted results; Proposition 6 is numerical. A certified lower bound for the Thurston distance would need certified lengths of closed geodesics on $h_P$ and, beyond that, control of the discretisation error of the uniformisation.

*A third twin.* The edges of $P$ form a trivalent ribbon graph with eight boundary cycles of length nine. Brooks and Makover [9] turn it into a hyperbolic surface of genus 3 with eight cusps; compactified, it is a third canonical point next to $h_{\mathrm{cp}}$ and $h_P$. Comparing the three Laplace spectra is natural.

*Symmetric periods.* For $K$ the action of $\tau$ on $H_1(S;\mathbb{R})$ has character $(6,0,-2,0)$, so a symmetric conformal embedding in Taimanov's spinor representation [30] has five rather than eighteen periods to close.

## 12. Computations and code

All computations use Python 3.12 with NumPy, SciPy, JAX, python-flint, CVXPY and SymPy, together with Singular [15] and msolve [5] (Singular 4.4.1p5, msolve commit `a18d0de` of 14 October 2026); all Python package versions are pinned in the repository. The large searches ran as job arrays on a Slurm cluster of three nodes with AMD EPYC 9554 processors, 128 CPUs and 384 to 768 GB of memory each; everything else takes seconds to minutes on a laptop. Table 6 lists, for every result, the script that produces it and its cost. The repository is archived at Zenodo [29]; its README gives the exact calls in order, from Mizhaev's coordinates, which are embedded verbatim, to the last figure. Every figure of this paper is produced by a script in the repository, and an interactive viewer shows $P$ next to surfaces conformal to $h_P$.

Table 6. Every result with the script that produces it and its cost. Paths are relative to the folder `conformal-twin-lab-ipc` of the repository. *Allocated CPU time of the three sweeps; the other costs are measured compute time.

| result | script | cost |
|---|---|---|
| triangulation $K$, circle packing $h_{\mathrm{cp}}$ | `src/experiment.py`, `src/develop.py` | seconds |
| Theorem 3 (certified radii) | `src/certify_radii.py` | seconds |
| Proposition 6, Table 3 | `src/wpc/wp_c.py`, `src/wpc/roundtrip.py` | minutes |
| Table 4, Section A | `src/pants/cuffs.py`, `src/ribbon.py` | minutes |
| Proposition 8, Theorem 9, Remark 10 | `src/alex/stress.py` | 25 s |
| Theorem 11, exact | `src/symm/msolve_input.py`, msolve; Singular | 34 s; 19 min |
| Theorem 11, homotopy check | `src/symm/run_symm.py` ($5 \times 131{,}072$ paths) | 6 CPU h |
| random search (6.4 million starts) | `src/alex/search.py` | 50 CPU h |
| embedding searches (Section 10) | `src/conformal_embed.py` | 1,343 CPU h* |
| Figures 1 to 8 | `visual/fig_*.py` | minutes |

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## Disclosure statement

The author reports there are no competing interests to declare.

## Declaration of generative AI use

During the preparation of this work the author used Claude Opus 5.5 (Anthropic) to edit the language of the manuscript, to help draft and revise its text, and to write and debug the code for the computations. The author ran all computations himself, checked every result, proof and program, and takes full responsibility for the content of this article.

## Data availability statement

All code, input data and results supporting this article are openly available at `https://github.com/Sultanow/cone-operator-lab` (folder `conformal-twin-lab-ipc`) and archived at Zenodo [29], doi:10.5281/zenodo.23423742.

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## Appendix A. The ribbon graph of $P$

For each vertex $v$ of $P$, in Mizhaev's numbering, the three neighbours in the cyclic order induced by the orientation of the surface. Following the rule "from the edge $uv$ continue with the edge $vw$, where $w$ follows $u$ in the cyclic order at $v$" traces the eight nonagons. The script `src/ribbon.py` computes the table and checks both statements.

| $v$ | order | $v$ | order | $v$ | order | $v$ | order |
|---|---|---|---|---|---|---|---|
| 1 | $(2,5,12)$ | 7 | $(5,18,14)$ | 13 | $(8,22,21)$ | 19 | $(6,23,12)$ |
| 2 | $(1,11,6)$ | 8 | $(3,13,5)$ | 14 | $(7,24,15)$ | 20 | $(12,21,24)$ |
| 3 | $(4,8,17)$ | 9 | $(4,23,15)$ | 15 | $(9,14,11)$ | 21 | $(10,13,20)$ |
| 4 | $(3,16,9)$ | 10 | $(6,21,16)$ | 16 | $(4,22,10)$ | 22 | $(13,18,16)$ |
| 5 | $(1,7,8)$ | 11 | $(2,18,15)$ | 17 | $(3,23,24)$ | 23 | $(9,19,17)$ |
| 6 | $(2,19,10)$ | 12 | $(1,19,20)$ | 18 | $(7,22,11)$ | 24 | $(14,20,17)$ |

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